KARDASHEV

Helping humanity move a step further, starting with lunar helium-3.

READ THE WHITE PAPER

PREAMBLE

NOTE TO THE READER

This is a white paper, not a pitch. It argues a position, shows the arithmetic behind it, and states what is not yet known. It is written for people who will check the numbers: engineers, scientists, and investors who have seen enough space presentations to distrust adjectives.

Three conventions apply throughout.

Ranges, not points. Where a value is measured, it is cited. Where it is modelled, it is labelled. Where it is assumed, the assumption is visible. A single number without a status is a defect. Section 6 explains why this document, and this programme, move at startup speed.

The unit is the kilogram of delivered product. Not the tonne of regolith moved, not the watt installed, not the headline resource. A mine is a machine for converting capital and energy into a saleable kilogram. That conversion is the only thing that matters.

Failures stay in the record. The programme behind this paper keeps its own failed designs. One of them is described in section 5. A plan that cannot survive its own falsification is not a plan.

All numbered sections have had internal fresh-context reviews (requirement fit, citation screening, prose, and a technical pass) on 12–13 November 2026. The document as a whole has not been independently verified, peer-reviewed, or checked against hardware. It is a working document, not an offer or a solicitation. The scalar-check appendix arithmetic is published with the source at the project repository: https://github.com/amarionfred/kardashev

. . .

SECTION 1 OF 11

THE LADDER

Civilisation climbs by converting energy into capability. The next rung is not a larger rocket. It is a first off-Earth industrial base.

In 1964, Nikolai Kardashev proposed a scale for civilisations based on the power they can command. Type I uses the energy incident on its planet. Type II uses the output of its star. Type III uses the output of its galaxy. The scale is logarithmic; in Sagan's version, each type is ten orders of magnitude above the last. The scale therefore measures a change in kind, not degree.

The numbers depend on the convention [1][2]. Kardashev's original Type I threshold was about 4×10^{12} W. Carl Sagan's logarithmic version, which is the one in common use, puts Type I at 10^{16} W [1][2]. Humanity's primary energy use is about 620 EJ per year, or 2×10^{13} W [3]. On Sagan's scale we are at 0.73. Reaching Type I requires roughly five hundred times the power we currently use.

Five hundred is a large number, but it is not the interesting one. The interesting one is that energy is not the only input. A civilisation that wants five hundred times the power must build five hundred times the machines, and those machines must come from somewhere. Earth's biosphere, its ore bodies, and its launch capacity are finite. At some point the mass fraction that must be lifted out of Earth's gravity well becomes the binding constraint, not the fuel.

The Moon is the first place where that constraint can be tested. It is close enough for a return trip measured in days. It has no atmosphere, so a launch from its surface does not pay the same penalty as a launch from Earth. It has no global magnetic field, so the solar wind has reached its surface for most of its history. And it has no biosphere to protect, which means industrial experiments there are not automatically environmental questions.

The destination, stated plainly, is an industrial arc that moves the Moon from a visited outpost to a worked site. The reasoning, set out in the 2017 video essay Industrializing the Moon, runs as follows. The Moon's composition is close to Earth's crust: silicon, oxygen, aluminium, iron and, in its mare basalts, titanium, plus water ice in shadowed polar craters. Shipping mass from the lunar surface to Earth orbit costs less than lifting the same mass from Earth's surface. The Moon is already, in that account's phrase, a warehouse pre-positioned next to the expansion front: raw mass and fuel for expanded space operations. Once ore handling and fuel production exist, the same feedstock and the same base can tow O'Neill-class rotating habitats into orbit instead of down into Earth's gravity well. The same account's construction ballpark is about a million person-years per habitat, with centuries to run in, starting from modest thousand-person installations [26].

That framing is the motive for the company, and it is a vision, not a schedule. This paper does not claim any of that is near. It claims that the first rung is a measured kilogram of helium-3, sold, and that everything above it is earned, one gate at a time.

The industrial case is not that the Moon is closer than Mars. It is that the mass budget compounds. Every tonne of structure, propellant, or shielding made on the Moon is a tonne not lifted from Earth. Every tonne not lifted is propellant not burned, and every kilogram of propellant not burned is mass available for something else. The Moon's vacuum also removes the cost of vacuum chambers, and its shallow gravity well makes return cheap. Those advantages are real, but they compound only once a base exists. The first industry has to pay for the base before the base can pay for anything else.

None of this makes the Moon a destination. It makes it a test. The first lunar industry will be judged on the same terms as any terrestrial one: mass in, product out, energy per kilogram, capital per kilogram, and time.

The rest of this paper is about one product, chosen because it is the best first test of that chain.

. . .

THE RESERVE

The Moon is not short of helium-3. It is short of concentration.

Helium-3 is a light isotope of helium with one proton and two neutrons. It is primordial, produced in the first minutes after the Big Bang and in stellar interiors. Earth has very little of it: the atmosphere's helium is about one part per million helium-3, and the isotope is lost to space over geological time. The Moon has no global

magnetic field and almost no atmosphere, so the solar wind, which carries helium-3 at a few hundred parts per million of its helium, has struck its surface directly for most of its history. The ions implant into the top tens of nanometres of regolith grains, and they stay there.

The result is a resource with a strange shape. The total inventory is large. Model-based estimates put roughly 6.5×10^{8} kg in the upper regolith [5]. That is hundreds of thousands of tonnes of a material whose terrestrial price is measured in millions of dollars per kilogram. The estimate follows an orbital abundance model rather than any single crop of samples; it is quoted as a model, not a measurement, and it is theirs, not re-derived here.

The concentration is the problem. Measured samples span 2.4 to 26 parts per billion by mass. Apollo 11 sample 10084, the bulk sample collected by Armstrong, averaged 11.8 ppb across analyses, with a range of 9.22 to 17.9 ppb [6][7]. Two Apollo 17 samples in the programme's design basis measure 4.66 and 11.21 ppb [8]. The highest values correlate with ilmenite and titanium-rich mare basalts, because ilmenite retains implanted helium better than other minerals [9][10]. That correlation is why the candidate sites are a short list of titanium-rich maria rather than the whole Moon.

QUANTITY	VALUE	STATUS
He-3 grade, measured samples	2.4–26 ppb	Measured, sample-specific
Apollo 11 sample 10084	11.8 ppb (9.22–17.9)	Measured, sample-specific
Apollo 17 72501,27	4.66 ppb	Measured, sample-specific
Apollo 17 70011,18	11.21 ppb	Measured, sample-specific
He-3/He-4 ratio, lunar	$2.7–2.9\times10^{-4}$	Derived from assays [6][7]
He-3/He-4 ratio, terrestrial	1.4×10^{-6}	Reference value
Global inventory, upper regolith	~6.5×10^{8} kg	Model estimate

Two conclusions follow, and they pull in opposite directions.

The first is that the resource is real. The measurements are not speculative; they come from returned samples, and the mechanism is understood. A mine would not be chasing a phantom.

The second is that a resource map is not a mine. The global inventory is a model. The grade at any specific site is an interpolation between sample points and orbital remote sensing, not a measured assay. The difference between 5 and 15 ppb is a factor of three in the size of every machine downstream. So the first payload in the sequence is a prospecting instrument, not an excavator.

The programme's design basis treats the grade as an unknown with a range, not a fact. That is the correct status. Anyone who tells you the grade of their mine before they have measured it is selling something other than a mine.

THE ARITHMETIC

Heat is cheap on the Moon. Mass is not. The machine is a heat exchanger with a mining problem attached.

The extraction chain is simple to state and hard to execute. Regolith is excavated, sorted, heated, and the evolved gas is captured, cleaned, and separated into helium-3 and helium-4. Each step has a loss. The product is the product of all of them.

```
regolith → excavate → sort → heat → release → capture → clean → separate → store → deliver
             ↑          ↑       ↑        ↑         ↑        ↑         ↑
           losses     losses  losses   losses    losses   losses    losses
```

The first number is the feed ratio. At 10 ppb and 100% recovery, one kilogram of helium-3 is implanted across 100,000 tonnes of regolith. At 50% overall recovery, it is 200,000 tonnes. At 5 ppb, it is 400,000 tonnes. The ratio is the mine.

The second number is the heat. The programme has checked a published heat-capacity fit for the FJS-1 lunar soil simulant over its measured range, 313 to 570 K. The integral is 226,744 J per kilogram, confirmed by two numerical routes to a relative difference of 3×10^{-9}; both routes integrate the same published fit, so this checks the arithmetic, not the fit itself [11][12]. Heating to 800 °C, where published release experiments report about 95% of implanted helium released [7], roughly triples that figure; the extension above 570 K is an assumption, and it is labelled as one.

Put the two together. At 10 ppb, 50% overall recovery, and roughly 0.5 to 0.7 MJ per kilogram of feed, one gram of helium-3 costs 100 to 140 GJ of heat. That is 28 to 39 MWh. With 85% recuperation, a figure claimed by the published Mark-III extraction concept in Olson 2021 [7], the net figure falls to 4 to 6 MWh per gram.

CASE	FEED PER GRAM	HEAT PER GRAM	NET HEAT AT 85% RECUPERATION
10 ppb, 50% recovery	200 t	100–140 GJ	4–6 MWh
10 ppb, 30% recovery	333 t	165–235 GJ	7–10 MWh
5 ppb, 50% recovery	400 t	200–280 GJ	8–12 MWh

The energy is not the obstacle. At a generous $100 per MWh, 30 MWh is $3,000 per gram, or 15% of a $20,000 per gram sale price. With recuperation it is about 2%. The thermodynamics are affordable. The capital is not yet.

The third number is the night. The lunar day is 29.5 Earth days, so the night is 14.75 days long. Equatorial surface temperatures reach about 400 K at noon and fall below 100 K before dawn, so the thermal cycle is not a small perturbation [24]. A plant that runs on solar power must either stop, store, or find a site with near-continuous illumination. Stopping halves the calendar output. Storing is worse: a 1.4 MW plant needs about 500 MWh to run through the night, which at 250 Wh/kg is roughly 2,000 tonnes of batteries. That is an enormous

mass item, plausibly exceeding the plant itself (indicative; no plant exists to compare against). A thermal store is lighter for heat than a battery is for electricity, but it is still a mass item, and it has its own losses.

The fourth number is the separation. Lunar helium is isotopically enriched relative to terrestrial helium by a factor of about 200, because it comes from the Sun rather than from radioactive decay in the Earth. That is a real advantage, not a solution. Helium-3 boils at 3.19 K and helium-4 at 4.22 K. Separating them at 99% purity requires a cryogenic distillation column or a quantum-sieving process (preferential low-temperature adsorption of the lighter isotope) operating near 2 K, with the refrigeration penalty that implies. The programme's design basis carries the separation energy as an unknown, not a number, because no public measurement supports a defensible value for a lunar plant.

The fifth number is heat rejection. A plant that consumes megawatts must reject megawatts. At 300 K, a black body radiates about 460 W per square metre; with emissivity 0.9 and two sides, roughly 800 W per square metre. One megawatt therefore needs about 1,200 to 2,200 square metres of radiator, before dust and view-factor penalties. Heat rejection is not a detail. It is a structure.

The arithmetic does not say the mine is impossible. It says where the difficulty lives: in the mass of the machine, the length of the night, and the fraction of the gas that survives the chain. Those are engineering problems, and they are the ones the programme is built to solve.

. . .

THE MACHINE

A mine that can survive the night is a different machine from a mine that works for fourteen days.

The programme has carried four extraction routes as explicit alternatives, not as a favourite and three straw men.

1. **Thermal reference.** Heat selected feed and capture the evolved gas, with heat recovery from the processed solids to the incoming feed. This is the most transparent published architecture, and the baseline against which the others are judged.
2. **Warm preheat.** Recover heat from the gas and the solids at a lower temperature, reducing the radiative loss that dominates at high temperature. The trade is residence time and release fraction.
3. **Mechanical release.** Agitate or mill the regolith to release gas without heating the bulk. Published simulant experiments removed 70% and 96% of implanted helium-4 at two small flow rates without heating above room temperature [7]. The experiment used helium-4 in a simulant, not helium-3 in lunar regolith, and it measured retained gas, not captured product. It is a lead, not a result.
4. **Hybrid.** Capture the early mechanical release, then heat the residual feed. This is a candidate, not a claimed improvement. It adds equipment mass and failure points, and the programme treats those as costs to be weighed, not details to be waved away.

The architecture that follows from the arithmetic is modular. A single 100 tonne per hour plant produces about 0.5 grams per hour at 10 ppb and 50% recovery, or 4.4 kilograms per year if it runs continuously. A 10 tonne per hour unit produces a tenth of that. The smaller unit is easier to launch, easier to test, easier to replace, and easier

to sell into a market measured in kilograms. A fleet of small units also fails gracefully: one jammed excavator does not stop the mine.

This is not a claim that small is always better. It is a claim that the first unit should be expendable, and that the fleet architecture should be chosen after the first unit has measured the things a model cannot.

The programme has built a parameterised design basis for this machine [13]. It contains 157 traced quantities (F1 design-basis records), each labelled as measured, sourced, assumed, or unknown, and each tied to a source or an explicit assumption. It has two candidate architecture allocations with common interfaces, CAD/STEP assemblies whose pinned snapshots hash-verify byte-identical (716 of 716) at file scope, and a finite-cycle resource ledger that accounts for power, heat, storage, service, and time. Geometry-to-parameter binding remains unproven and is deferred to F3. It does not have a demonstrated machine. The distinction matters, and the paper keeps it.

What the design basis has already produced is a list of the things that break. The 256 K cold-feed case falls outside the measured heat-capacity range, so the low-temperature relation is an open gap, not an extrapolation. A stowed equipment recess leaves 12 unwaived penetrations at the current envelope. A funded design closure for one part of the extraction chain was falsified by a deterministic simulation at 0.263 s of model time (see section 5). None of these is a surprise in a first design. All of them are recorded, owned, and carried into the next phase rather than smoothed out of the presentation.

That is the machine's real state: an architecture with a traceable basis, a set of named defects, and no hardware. A white paper that claimed more would be overclaiming.

. . .

THE LEDGER

The programme keeps the failures. The failures are the reason to trust the process.

A technical claim has four possible states. It can be measured, modelled, assumed, or unknown. Space-resource presentations often collapse the last three into the first. This paper does not.

CLAIM	STATUS	WHAT WOULD SETTLE IT
He-3 grade at a candidate site	Unknown; sample range 2.4–26 ppb	A prospecting payload with a calibrated gas assay
Release fraction at 500–800 °C	Measured on Apollo sample 10084; not reproduced	Bench test on lunar-derived or well-characterised simulant
Capture efficiency	Unknown	Bench test with a sealed gas train and mass balance
Heat recovery	Modelled at 85% in a published concept	Measured recuperation on a sub-scale exchanger
Separation energy at 99% purity	Unknown	Cryogenic column test with measured power
Night survival	Modelled	Thermal-vacuum test of the storage and radiator
Dust wear on seals and bearings	Unknown	Long-duration simulant test
Robotic service	Requirement, not design	Executable service procedure with a robot
Cost per kilogram	Unknown; published estimates $20–100M/kg [4]	A costed pilot with real supplier quotes

The method behind the table is deliberately boring. One work item per defect. A written question, a resource budget, a checkpoint, and a stop condition. Three attempts per defect, no silent renaming, no widening of tolerances to make a run pass. An independent reviewer who does not edit the producer's files. A status record that separates a passed job from an accepted result.

The programme's own failure is the best evidence that the method is real. A funded design closure for part of the extraction chain was simulated deterministically and failed at 0.263 s of model time. The contract that defined it was not rewritten to save it. The failure was recorded with its time, its inputs, and its reviewer, and the design was carried forward as an open defect into the next phase. A reader can disagree with the design. A reader cannot say the failure was hidden.

This is slower than a pitch deck. It is also the only way to know which numbers are real.

THE EDGE

AI is why this programme's calendar closes. Here is the mechanism, not the motto.

The incumbent way to buy certainty in aerospace is headcount: specialists to read the literature, CAD operators to keep assemblies in sync with a parameter model, simulation teams to falsify designs, reviewers to catch what the simulation missed, writers to describe it. The failure modes of that method are well known: the coordination cost scales with the team, and the loop from idea to falsification takes months. The programme behind this paper runs the same workflows with AI in the loop. The loop is short, and its cost does not scale with headcount.

Every workflow in this programme runs with AI in it today:

WORKFLOW	WHAT AI DOES	EVIDENCE
Research	Reads the primary literature, keeps it cited and dated	The bibliography of this paper, every number labelled measured, sourced, modelled, or unknown
Design basis	Traces quantities through a parameterised model, labels each	157 traced quantities, F1-BASIS-C02
Architecture	Generates and compares candidate allocations; pinned CAD/STEP snapshots hash-verified	716 of 716, file scope (F2) [13]
Verification	Deterministic simulation falsifies designs before hardware exists	A funded thermal closure failed at 0.263 s of model time
Review	Fresh-context AI reviewers read the work, edit nothing, file findings	This paper's technical review, 12 November 2026, including a 1,000× break-even error it caught before any human read the draft
This document	Researched, drafted, reviewed and arithmetically checked in days	Version 1.0, 12 November 2026

The benefit is multiplicative, not additive. Each gate is cheaper than the gate before it because the machinery — the traceability, the reconciliation, the simulation, the review — is built once and reused. A simulated falsification costs a fraction of a bench test. A bench test costs a fraction of a flight. The same programme, run the incumbent way, prices the loop in person-months per gate; run this way, the loop closes in days.

The human hours go to judgment: what to build, what to stop, what to ask next. None of this removes physics. Dust does not care about the token budget. A simulation that runs is not a machine that works, and an AI review is a label, not a proof: the 157 quantities are honest because each is tied to a source or an explicit assumption, not because a model produced them. Where this paper says unknown, it means nothing has been measured. The AI edge removes the bottleneck between those measurements, and calendar time lives in that bottleneck.

That is the edge. It compounds, it is checkable, and it is how this machine gets measured on a startup's budget.

. . .

SECTION 7 OF 11

THE BUYER

The price of helium-3 is high because the supply is small, not because the demand is large.

This is the most important economic fact in the paper, and it cuts against the usual lunar mining story.

Global production of helium-3 is under four kilograms a year [4]. The United States produces about one kilogram a year [14], almost all of it as a by-product of tritium decay at the Savannah River Site. United States federal demand, after a decade of substitution and recycling, is below 6,000 litres a year, about 0.8 kilograms [14][15]. The price is quoted around $20 million per kilogram, with small-lot quotes ranging from $15,000 to $19,000 per gram [4][16].

A plant producing forty kilograms a year would not find forty times the customers. It would find a lower price. The demand curve, not the geology, sets the ceiling on the first mine. That is why the sequence in section 8 starts with a prospecting payload and a bench unit, not a hundred-tonne excavator.

The demand has three parts, and they are not equally durable.

Quantum cooling. Dilution refrigerators use a helium-3/helium-4 mixture to reach millikelvin temperatures. Superconducting qubits need that cooling. Quantum computing is the fastest-growing source of demand, and it is where the first large commercial contracts for lunar helium-3 have appeared [4]. It is also the segment most exposed to substitution: researchers are actively engineering around helium-3 because the supply is scarce and the price is high.

Neutron detection. Helium-3 was the standard fill gas for neutron detectors used in port and border security. Demand spiked after 2001, drained the United States stockpile, and forced rationing [15]. The shortage accelerated alternatives: lithium-6 and boron-10 detectors now meet the same requirement without helium-3 [17]. This segment is unlikely to grow.

Fusion. The deuterium-helium-3 reaction is aneutronic in its main branch, which means less wall damage and the possibility of direct energy conversion. It also ignites at roughly 700 million degrees Celsius, about four times the deuterium-tritium temperature, and no machine is close to it [18]. Only a few fusion developers plan to buy helium-3 on the open market; most plan to breed their own or use deuterium-tritium. Fusion is an option on the future, not a customer today.

The honest summary is that helium-3 is a niche isotope market with a strategic option on fusion. The niche is real: the price is real, the contracts are real, and the supply is genuinely constrained. The option is real too, but it is an option, and a company that needs fusion demand to survive will not survive long enough to see it.

What follows is the case for sizing the first plant to the niche, and building the industrial base that makes the next product possible. The same feed handling, power, and thermal infrastructure that heats regolith for

helium-3 can support oxygen, metals, and sintered structural material. Those products have larger markets and lower prices. They are the base. Helium-3 is the margin that pays for the first unit.

The comparison is worth stating plainly. At 10 ppb and 50% recovery, a tonne of regolith yields 5 milligrams of helium-3, worth about $100 at $20 million per kilogram. A tonne of regolith at 10% aluminium and 50% recovery, an optimistic assumption used for illustration, yields 50 kilograms of aluminium, worth about $198 at $3.97 per kilogram [19]. By gross revenue per tonne of feed, aluminium wins. By revenue per kilogram of product, helium-3 wins by a factor of five million. The first metric governs the size of the machine. The second governs the size of the market. The arithmetic is in Appendix B. A mine has to satisfy both, and the programme's model is built to find out whether any configuration does.

That is the question the first plant is designed to answer. A white paper cannot answer it.

. . .

THE SEQUENCE

The first machine should be small enough to fail.

The programme is organised as a sequence of gates. Each gate has a question, a measurement, and a kill criterion. A gate that cannot be failed is not a gate.

Gate 0: Prospect. A small payload measures helium-3 and helium-4 at a candidate site, with a calibrated gas assay and a known sampling depth. The question is whether the grade is above 5 ppb, and whether it is spatially uniform enough to mine. The kill criterion is a measured grade below 5 ppb at the best accessible site, which would make the current architecture uneconomic and force a re-selection of product or site. A prospecting payload is of the order of tens of kilograms, not tonnes (an estimate; no payload design exists yet). It is the cheapest way to retire the largest unknown.

Gate 1: Bench. A laboratory unit processes simulant and, where available, lunar-derived feed. It measures release fraction versus temperature and time, capture efficiency, heat recovery, and the mass balance of the gas train. The question is whether 50% of the implanted helium can be recovered at 500 °C with at least 80% recuperation (the published Mark-III concept claims 85%). The kill criterion is a measured release below 30% at 800 °C, which would break the thermal route and move the programme to mechanical or hybrid extraction.

Gate 2: Pilot. A sub-scale unit, on the Moon, operates for one lunar day. It measures gas, power, dust, and thermal performance in the real environment. The question is whether the machine survives the night and whether the measured output matches the model within a factor of two. The kill criterion is a measured output below one-fifth of the model, which would mean the loss chain is not understood.

Gate 3: Production. A small fleet produces a few kilograms a year and sells them into the niche market. The question is whether the unit economics close at the contract price. The kill criterion is a cost per kilogram above the contract price after two years of operation.

Gate 4: Base. The same fleet produces bulk products for lunar use: oxygen, metals, and structural material. The question is whether the marginal cost of those products is below the cost of delivering them from Earth. This is

where the industrial arc starts paying for itself: the industrial base becomes the business, and helium-3 becomes the margin rather than the product.

The sequence is deliberately front-loaded with measurement, and it is achievable on a startup calendar precisely because the loop in section 6 makes each gate cheap to close. A simple cost model says the first plant is dominated by launch mass: at $1 million per kilogram delivered to the lunar surface and $20 million per kilogram of product, every tonne of hardware must produce fifty kilograms over its life to break even. At $100,000 per kilogram, the requirement falls to five kilograms per tonne. That single ratio is a concrete design driver, because it tells the engineer what to minimise.

LAUNCH PRICE	PRODUCT NEEDED PER TONNE OF HARDWARE, OVER LIFE
$1,000,000/kg	50 kg
$500,000/kg	25 kg
$100,000/kg	5 kg

No public data establishes a lunar plant's mass per unit of throughput. Hence the sequence starts with a pilot. The pilot measures the mass. The mass sets the launch requirement. The launch requirement sets the capital. The capital sets the price. The price meets the demand ceiling. The chain is closed, and every link is a number.

. . .

SECTION 9 OF 11

THE LANDSCAPE

Nobody has flown the assay. That is the whole game.

Interlune, founded by Rob Meyerson after his time as president of Blue Origin, is the most serious announced player in this field. Its CTO Gary Lai was chief architect at Blue Origin; its strategic advisor is Harrison Schmitt, Apollo 17 astronaut and geologist. Its record is real: a full-scale Vermeer excavator ran in Iowa in 2025, pure helium-3 has been produced from domestic helium by cryogenic processing, and it claims nearly $500 million in binding helium-3 purchase agreements with Maybell Quantum, Bluefors and the U.S. Department of Energy — its own accounting, not audited — named for the quantum-cooling segment, not a mass market [25].

One fact frames the whole field: no kilogram of off-world helium-3 has flown, no site on the Moon has returned an assay, and no announced commercial prospect is anything other than a forward instrument curve standing on sample interpolation. The buyers are betting. Interlune is building. The unknown itself is still unmeasured.

This programme sits in that gap deliberately, and it is the entire reason the company exists in this shape.

The edge, stated plainly.

- Nobody has flown an in-situ prospecting assay. Every announced prospect from here, Interlune included, prices against the number our Gate 0 is built to produce. The first flown assay becomes the ground truth; every miner, every contractor and every buyer prices from it afterwards.
- Our engineering loop runs at AI speed, not team speed. The reading, the CAD reconciliation, the simulation, the review and the drafting close in days instead of months, and they are published and checkable in the public repository. That is not a slogan. It is why this paper exists and exists this early.
- The failure ledger is published, not buried. An error found before a reader sees it is cheaper than a design frozen behind a rework crisis, and far cheaper than a mission lost.
- We burn capital in gate-sized increments with stated kill criteria. Each dollar buys one question answered or one reason to stop. That is a structurally lower burn than a company building an excavator in Iowa and an assay in parallel.
- While Interlune owns the purchase orders, we own the unknown they all depend on. Their business case inherits our measurement the day it flies.

The competitive read, without varnish. Interlune has the team, the capital and the customers; it is a serious company doing hard things in public. This programme is doing a harder thing on a quieter shoestring: proving that a measured kilogram can be obtained at all. If you believe in the contracted buyers clearing before any mine is real, Interlune is the ticket. If you believe no one gets to the moon without the assay — every one of them, us included — then the team that flies it first owns the most valuable piece of information in lunar industry. That is us.

China is the other gravity well. Beijing Research Institute of Uranium Geology led the 2023 noble-gas measurements in Chang'e-5 low-titanium basalts — trapped 3He/4He less fractionated than Apollo soils [27]. No He-3 concentration is published. A new commercial effort, Starpath, announced in February 2026 with a hyperspectral 3He-prospecting lander passed for feasibility review in September 2026 — surveying, not mining, and no flown hardware. Ouyang Ziyuan's "10,000 years of energy" line is rhetoric, not a plan; the ILRS programme lists no helium-3 goal [27]. The measurement question is now the contested ground, and whoever answers it first will own the argument that follows.

. . .

THE QUESTIONS

Ten questions a technical reader should ask, with the answers this paper can defend.

Why helium-3 rather than water or metals? Because it is the only lunar material whose price per delivered kilogram is high enough for a small plant to be revenue-relevant. Water and metals have larger markets and lower prices; they become attractive when the industrial base exists to make them cheaply. Helium-3 is the bootstrap, not the destination.

Isn't the market too small? Yes, for a large plant. The market is about one to four kilograms a year [4][14][15] at current demand and supply. Hence the first plant is small, and the business case rests on the niche plus the industrial base, not on fusion demand.

Why would anyone buy lunar helium-3 when terrestrial supply exists? Because the terrestrial supply is constrained by tritium decay and by the cost of separation from natural gas. A second source with a different cost structure is strategically valuable, and the first commercial contracts show buyers will pay for it. Whether they will pay enough at scale is unproven.

Can the Moon supply it at a competitive cost? The physics allows it. The published estimates of $20–100 million per kilogram bracket the current price, which means the answer depends on launch mass, life, and recovery. The programme does not claim a cost. It claims a method for measuring one.

What about the 14-day night? It is a design variable. A plant can stop, store, or site for illumination. Each choice has a mass and output penalty. The design basis models all three, and the pilot measures the one selected.

Why not use nuclear power? It is a candidate for later phases. Solar is the primary source because it is abundant, scalable, and does not require a reactor qualification. A reactor would remove the night constraint at the cost of mass, licensing, and thermal rejection. The trade-off remains open; the design basis models both paths.

Is the separation the hard part? It is one of the hard parts. The lunar helium is enriched about 200 times relative to terrestrial helium, which helps. Reaching 99% purity at 2 K still requires a cryogenic plant with a measured power budget. No public measurement supports a lunar-scale number, so the programme carries it as an unknown.

What is the legal status? The Outer Space Treaty prohibits national appropriation of the Moon but does not settle whether a private party may extract and sell a resource [20]. The Moon Agreement, which treats resources as common heritage, has 17 parties (Saudi Arabia withdrew, effective 2024) and binds none of the major spacefaring states [21]. The United States, Luxembourg, the UAE, and Japan have enacted domestic frameworks recognising extracted resources as property [22]. The Artemis Accords, signed by 78 states as of October 2026, state that resource extraction does not inherently constitute national appropriation [23]. The practical position is that extraction is probably lawful for an operator whose state recognises it, and that the first serious dispute will be political, not technical.

What happens if fusion never arrives? The niche market still exists, the industrial base still has value for lunar operations, and the option expires worthless. That is an acceptable outcome if the first plant is sized to the niche. It is not acceptable if the plan requires fusion demand to repay the capital.

What is the biggest risk? The biggest risk is not geology, and not thermodynamics. It is that the loss chain, the sequence of small efficiencies from excavation to storage, multiplies to a recovery so low that the machine cannot pay for its own mass. The bench and pilot gates exist for that reason.

. . .

SECTION 11 OF 11

WHAT THIS PAPER DOES NOT CLAIM

It does not claim that a lunar helium-3 mine exists. It does not claim a cost per kilogram. It does not claim a measured grade at any site. It does not claim that fusion will create demand. It does not claim that the current architecture is the best one. It does not claim that the programme's design basis is hardware.

It claims that the problem is real, that the arithmetic is checkable, that the first machine can be small, and that the sequence from prospecting to production is a sequence of measurements rather than promises.

. . .

REFERENCES

REFERENCES

[1] Kardashev, N. S. (1964). Transmission of information by extraterrestrial civilizations. Soviet Astronomy, 8, 217.

[2] Sagan, C. (1973). The Cosmic Connection. Anchor Press. Logarithmic scale: K = (log10 P − 6)/10.

[3] Energy Institute (2024). Statistical Review of World Energy 2024. Primary energy consumption, 620 EJ in 2023, ~1.96×10^{13} W.

[4] Nasr, M. (2026). Who Owns the Moon? A Case Study on Helium-3. Belfer Center for Science and International Affairs, 26 October 2026. https://www.belfercenter.org/research-analysis/who-owns-moon-case-study-helium-3

[5] Fa, W., & Jin, Y.-Q. (2007). Quantitative estimation of helium-3 spatial distribution in the lunar regolith layer. Icarus, 190, 15–23. https://doi.org/10.1016/j.icarus.2007.03.014

[6] Hintenberger, H., Weber, H., Voshage, H., Wänke, H., Begemann, F., & Wlotzka, F. (1970). Concentrations and isotopic abundances of the rare gases, hydrogen and nitrogen in Apollo 11 lunar fines. Proceedings of the Apollo 11 Lunar Science Conference, 1.

[7] Olson, A. D. S. (2021). Lunar Helium-3: Mining Concepts, Extraction Research, and Potential ISRU Synergies. AIAA ASCEND 2021. NASA NTRS 20210022801. https://ntrs.nasa.gov/citations/20210022801

[8] Internal design basis, F1-BASIS-C02. Apollo 17 sample assays 72501,27 and 70011,18: 4.656 and 11.209 ppb. Internal, versioned.

[9] Johnson, J. R., Swindle, T. D., & Lucey, P. G. (1999). Estimated solar wind-implanted helium-3 distribution on the Moon. Geophysical Research Letters, 26, 385–388. https://doi.org/10.1029/1998GL900305

[10] Crawford, I. A. (2015). Lunar resources: A review. Progress in Physical Geography, 39, 137–167. https://doi.org/10.1177/0309133314567585

[11] Nakayama, Y., Kudo, N., Kuroiwa, N., Endo, R., Ota, H., & Nishi, T. (2025). □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ [Specific heat of simulated lunar sand by temperature-modulated differential scanning calorimetry]. Experimental Mechanics, 25(1), 14–17. https://doi.org/10.11395/jjsem.25.14. The FJS-1 designation comes from the internal design basis, not the article metadata.

[12] Internal design basis, F1-BASIS-C01/A01 scalar check. 226,744.1 J/kg over 313–570 K, two numerical routes, relative difference 3.19×10^{-9}. Internal, versioned.

[13] Internal design basis (F1-BASIS-C02) and the F2 architecture study. Two candidate allocations; CAD/STEP pinned snapshots hash-verified 716/716 at file scope; geometry-to-parameter binding unproven, deferred to F3. Internal, versioned; closed at declared conditional scope, not accepted as hardware.

[14] US Department of Energy, Isotope Program. Supply and Demand of Helium-3. https://www.isotopes.gov/Supply-and-Demand-of-Helium-3

[15] US Government Accountability Office (2011). Managing Critical Isotopes: Weaknesses in DOE's Management of Helium-3 Delayed the Federal Response to a Critical Supply Shortage. GAO-11-472. https://www.gao.gov/products/gao-11-472

[16] Pulsar Helium. Helium-3 price range, $15,000–$19,000 per gram. https://pulsarhelium.com/he-hub/about-he/

[17] NIST (2018). Neutron detection method: long-sought workaround to helium shortage. https://www.nist.gov/news-events/news/2018/06/nist-neutron-detection-method-long-sought-workaround-helium-shortage

[18] Temmerman, G. D. (2021). The helium bubble: prospects for ³He-fuelled nuclear fusion. Joule, 5, 1312–1315. https://doi.org/10.1016/j.joule.2021.05.003

[19] USGS (2026). Mineral Commodity Summaries 2026: Aluminium and Titanium. https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-aluminum.pdf

[20] United Nations (1967). Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, including the Moon and Other Celestial Bodies. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/outerspacetreaty.html

[21] United Nations (1979). Agreement Governing the Activities of States on the Moon and Other Celestial Bodies. https://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/moon-agreement.html — 17 parties; Saudi Arabia withdrew, effective 5 January 2024 (18 was cited in earlier drafts; the withdrawal reduced the count).

[22] US Commercial Space Launch Competitiveness Act, H.R. 2262 (2015); Luxembourg Law of 20 July 2017; UAE Federal Law No. 12 of 2019; Japan Act on the Promotion of Business Activities for Exploring and Developing Space Resources (2021).

[23] NASA. The Artemis Accords; 78 signatories as of 28 October 2026 (Vietnam, the 78th, signed that date). https://www.nasa.gov/artemis-accords/. Moon Agreement comparison: [21].

[24] Vasavada, A. R., et al. (2012). Lunar equatorial surface temperatures and regolith properties from the Diviner Lunar Radiometer Experiment. Journal of Geophysical Research, 117, E00H18. https://doi.org/10.1029/2011JE003987

[25] Interlune Corporation. Sources: https://interlune.space/ (claims nearly $500M in binding helium-3 purchase agreements, Maybell Quantum and Bluefors, May 2025 DOE Isotope Program 3-litre purchase), https://interlune.space/about (Meyerson, Lai, Hornsby, Schmitt adviser), https://interlune.space/press-releases (NASA $6.9M lunar resources contract May 2026, Vermeer excavator partnership 2025–2026, pure helium-3 from domestic helium July 2026, U.S. Air Force SBIR Direct-to-

Phase II November 2025, helium-implanted simulant testing August 2026). Interlune's purchase figures are self-reported, not audited.

[27] Li, Y., et al. (2023). Noble-gas analyses of Chang'e-5 low-titanium basalts: high 4He/36Ar and 20Ne/36Ar, trapped helium less fractionated than Apollo soils. Communications Earth & Environment 4, 278. https://www.nature.com/articles/s43247-023-00937-9 — no published He-3 concentration from CE-5/CE-6 material exists as of November 2026. ILRS roadmap: https://www.cnsa.gov.cn/english/n6465652/n6760280/c6812168/content.html — no He-3 goal.

[26] Industrializing the Moon (2017, video essay). Vision framing for section 1. Popular exposition, not an engineering source. No technical claim in this paper depends on it.

. . .

APPENDIX A

UNITS AND CONVERSIONS

- 1 ppb by mass = 1×10^{-9} kg/kg = 1 mg per tonne.
- 1 tonne = 1,000 kg.
- 1 g of helium-3 at 10 ppb and 100% recovery requires 100 t of regolith; at 50% recovery, 200 t.
- 1 GJ = 277.8 kWh; 1 MWh = 3.6 GJ.
- 1 kg of helium-3 at $20 million/kg = $20,000 per gram.
- 1 litre of helium-3 gas at STP ≈ 0.135 g, so 7.4 litres ≈ 1 g.

APPENDIX B

THE ARITHMETIC, STEP BY STEP

1. Feed ratio: `m_feed = m_product / (grade × recovery)`.
2. Sensible heat: `Q = m_feed × ∫cp dT`. Over 313–570 K, the FJS-1 fit gives 226,744 J/kg [11][12].
3. Heat per gram: at 10 ppb and 50% recovery, 200 t × 0.5 MJ/kg = 100 GJ.
4. Net heat with recuperation: `Q_net = Q × (1 - r)`. At r = 0.85, 15 GJ = 4.2 MWh.
5. Night storage: `E = P × t_night`. At 1.4 MW and 1.275×10^{6} s, 1.78×10^{12} J = 495 MWh.
6. Break-even launch mass: `m_product / m_hardware = price_launch / price_product`. At $1M/kg and $20M/kg, the ratio is 0.05, i.e. 50 kg of product per tonne of hardware.
7. Aluminium comparison (illustrative, optimistic): 1 tonne of feed at 10% aluminium, 50% recovery → 50 kg Al ≈ $198 at $3.97/kg [19]; the same tonne yields 5 mg He-3 ≈ $100. Gross revenue per tonne favours aluminium; revenue per kilogram of product favours helium-3 by a factor of about 5 million.

STATUS OF THE NUMBERS

Measured: sample assays, heat-capacity fit within its range, surface temperatures, release fractions from Apollo sample 10084 [6][7]. Modelled: global inventory, plant mass, heat recovery. Assumed: site grade, overall recovery, launch price, plant life. Unknown: separation energy at 99% purity, capture efficiency, dust wear, robotic service life, cost per kilogram.

The paper's argument does not depend on any single unknown being favourable. It depends on the sequence that retires them in order of cost.